

DATA CHALLENGE

Diameter

Context

The database **inventory.xlsx** that you will use is not a fictional textbook example; it is a **real forest inventory** of a eucalyptus plantation.

Variable	Value in the plot
Branch	2
Plot	8
Age	7 years
Regime	Reform
Spacing	3 x 3 m
Clone	1501
Planting month	9
Place	1
Sampled plots	6 (numbered 1 to 6)
Trees in total	215

Dictionary of columns:

Column	Meaning
INSTALLMENT	Plot number (sampling unit) within the plot
ROW	Planting line within the plot (1 to 7)
TREE	Sequential number of the tree within the plot.
DAP	Diameter at chest height (cm)
HT	Total height (m) — note: not all trees have a measured total height.

Each plot has 7 rows of trees, with a variable number of trees per row (due to gaps/replanting in the plot). The DBH (diameter at breast height) was measured in all 215 trees. However, the height (HT) appears as 0 in most of the rows; this is not an error or a typo: it is a sampling decision that is part of the problem to be solved.

The problem

You have been (hypothetically) hired to audit the quality and summarize the diameter data for plot 8 before it goes into the official inventory report. The forestry manager wants answers to four questions:

1. Characterize each plot and the field, including its arithmetic mean diameter, quadratic mean diameter, standard deviation of diameters, largest and smallest diameter, and coefficient of variation of diameters.
2. Group the diameter data for each plot and the entire field into diameter classes. Use the respective standard deviations and ranges for the classes. Present tables with the grouped data showing the observed and cumulative frequency of individuals, in absolute values and percentages. How many individuals with a diameter greater than or equal to 15 centimeters can we expect in this field, considering that the field has hectares?

the

48.7

3. What is the basal area (m^2) and the basal area per hectare (m^2/ha) of each plot and of the plot as a whole?
4. Are the DBH data reliable? Are there any suspicious values (typing errors, out-of-standard measurements) that need to be investigated before calculating the baseline area? What statistical criteria did you use to decide what is "suspicious" (e.g., interquartile range, z-score, comparison with the distribution of the plot itself)?
5. What is the diameter distribution of the stand? Construct a DBH histogram (by plot and for the entire stand) and comment on what it reveals about the planting structure (uniformity, presence of dominated trees, etc.).
6. The spacing is 3×3 m. Based on this information and the tree/row count for each plot, estimate the area of each plot and the number of trees per hectare. Does this number make sense for a eucalyptus plantation with this spacing? Explain the assumptions made (the base does not provide the plot area—you will have to deduce it).

Delivery (GitHub)

- Commented script (R or Python) with all the steps above.
- README explaining the quality control routine adopted (the chosen outlier criterion and by what).
- Short report with the main figures and interpretation of the histogram.